

Seeing Beyond Darkness With SMART VISION WALLET

Smart vision wallet is a simple and efficient money counter and a counterfeit detector designed to be used by the visually-impaired and hearing-impaired community. Most machines use an LCD display, and are expensive and difficult to carry. Smart vision wallet is cheaper and comprises lighter and smaller components, reducing it to a size of an ordinary wallet—making it easy to carry

Dr S. Balakrishnan
is professor,
department
of information
technology, Sri
Krishna College
of Engineering
and Technology,
Coimbatore

Dr Susan Augustine
is associate
professor,
department
of information
technology, Sri
Krishna College
of Engineering
and Technology,
Coimbatore

Dr J.P. Ananth
is professor,
department of
computer science
and engineering,
Sri Krishna College
of Engineering
and Technology,
Coimbatore

As money has become extremely important for everyday life, the blind are at a big disadvantage. The visually-impaired community have no security when it comes to cash-related matters, and can be tricked easily. At times, they are even given false or wrong currency notes or the wrong sum total of currency. It is also easy to steal from them.

The Reserve Bank of India (RBI) in consideration for the blind community has introduced symbolic references in newer currency notes, such as Intaglio markings, which can be detected through touch. The main drawback of this feature is that not everyone can easily identify these symbols. The markings are introduced with the belief that everyone who is blind is sensitive to touch and can identify these.

How smart vision wallet helps

The smart vision wallet notifies the user regarding the sum of money counted as a sequence of vibrations and as voice output, which is intended for the hearing-impaired and the visually-impaired, respectively. Having such a device allows them to handle money with utmost confidence and independence.

There are many products available for the visually-impaired. Radio frequency identification (RFID) utilises electromagnetic fields to carefully exchange information. However, someone stand-

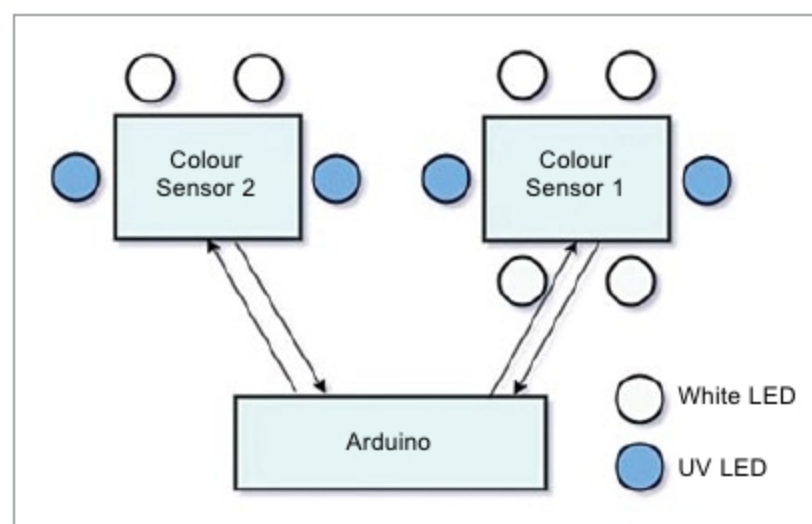


Fig. 1: Currency identification

ing behind a visually-impaired person in a supermarket queue could quietly extract the latter's information using silent technology.

A rechargeable smart wallet available in the market is a smart Bluetooth-powered wallet that shields the wallet from getting lost. Such options are great for the common people, but do not address the problems faced by the visually- and hearing-challenged people of the society.

What is smart vision wallet

Smart vision wallet consists of colour sensors, Bluetooth interface, ultraviolet LEDs, microprocessor and vibrator. The microprocessor analyses the data obtained from colour sensors to identify the denomination of the currency. Ultraviolet LEDs, in conjunction with the sensor and microcontroller (MCU), check the authenticity of the currency. After the processing